

Speech as an Alzheimer's Screen: Proof-of-Concept Findings

Can off-the-shelf speech analysis detect AD from ordinary audio - and how early? | Internal memo, July 2026

Bottom line

Using only open, uncontrolled YouTube audio and off-the-shelf features, a speaker-independent model detects Alzheimer's-related speech change at **AUC 0.62 on a locked, never-seen test set** - a modest but real signal. Just as important is what rigorous testing ruled **out**: a flashier model, and a striking 'signal-appears-a-decade-early' curve, both looked great in cross-validation and **failed to replicate** on held-out data. Honest read: the core signal is real but weak, and casual-audio early detection is **not yet demonstrated**. This de-risks the hypothesis cheaply and tells us exactly what to fix.

What we built

Data: DementiaNet - a free, longitudinal corpus of spontaneous speech from public figures with confirmed dementia and cognitively healthy age-matched controls, with clips tagged by years-before-diagnosis. We restricted the patient group to an Alzheimer's-enriched phenotype (excluded Parkinson's, Lewy body, primary progressive aphasia). Final set: **344 clips, 171 speakers** (115 dementia / 229 control).

Method (built for honesty, not for a big number): speaker-independent cross-validation, a locked 25% test set touched exactly once, and a documented confound audit - we caught the model exploiting **clip length** (an artifact of how videos were trimmed) and removed it. Features: voice-quality/prosody (openSMILE eGeMAPS) plus pause-timing measures.

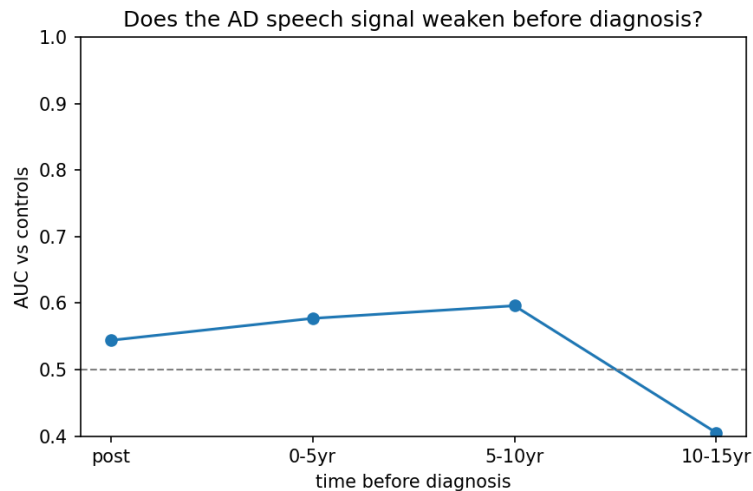
The rigor paid off: cross-validation alone would have misled us

Model	Dev cross-val	Held-out test
Logistic regression	0.57	0.61
SVM (RBF)	0.60	0.62
Gradient boosting	0.60	0.42 (overfit)

Gradient boosting had the best cross-validation score but **collapsed below chance on the held-out test** - it had memorised quirks of the training speakers. The regularised linear models generalised. We reject gradient boosting and adopt the linear model. **This is the locked test set doing its job**: catching an overfit that cross-validation hid.

Does the signal appear years before diagnosis? Not confirmed

We split patient clips by years-before-diagnosis and scored each slice against controls. A flexible model suggested a clean decline from diagnosis backwards - the exciting early-detection curve - but it **did not survive** the switch to the model that generalizes. Under honest evaluation the per-bin scores hover near chance with no reliable ordering (recent 0.56 vs earlier 0.51). So we do **not** claim a validated early-detection signal from this corpus: the striking gradient was an overfitting artifact. Confirming pre-symptomatic detection will need controlled tasks and clinical data (next steps).



On feature families, voice-quality/prosody carries what signal there is; pause timing is weak, and automatically-detected filled pauses (uh/um) add nothing on this corpus - a clean, documented negative result.

What this means for the venture

- **Core signal validated as real but modest.** ~0.62 on held-out data confirms speech carries AD information - but well below a usable screen, and early detection from casual audio is unproven on this data.
- **Rigor is the asset.** The pipeline killed two tempting false positives - an overfit model and a spurious early-detection curve - that a less careful team would have pitched. That discipline is what makes future claims credible to clinicians and regulators.
- **Clear path to a real signal:** gains will come from controlled elicitation tasks and biomarker-confirmed clinical audio, not from squeezing this corpus.

Honest limitations

- AUC ~0.62 is a proof-of-concept, **far from a deployable screen** (published clinical, task-controlled systems reach 0.77-0.93).
- Celebrity interview audio, edited and noisy; diagnoses are public-record, not biomarker-confirmed; per-time-bin and test samples are small, so numbers are trend-level, not precise.
- English-dominant; no demographic fairness testing yet.

Next steps

- Move to **controlled elicitation** (picture description, story recall) - the tasks behind the field's 0.77-0.93 results.
- Pilot **proprietary, biomarker-confirmed clinical audio** - the real moat.
- Test **deep acoustic embeddings** (wav2vec2 / HuBERT) vs the eGeMAPS baseline.
- Scope the regulatory path (Software-as-a-Medical-Device) for a speech screen.

Reproducible pipeline: speaker-independent CV, locked test split, confound removal, feature ablation, and longitudinal analysis - all in the project repo.